



MAE C163A/C263A
Kinematics of Robotic Systems

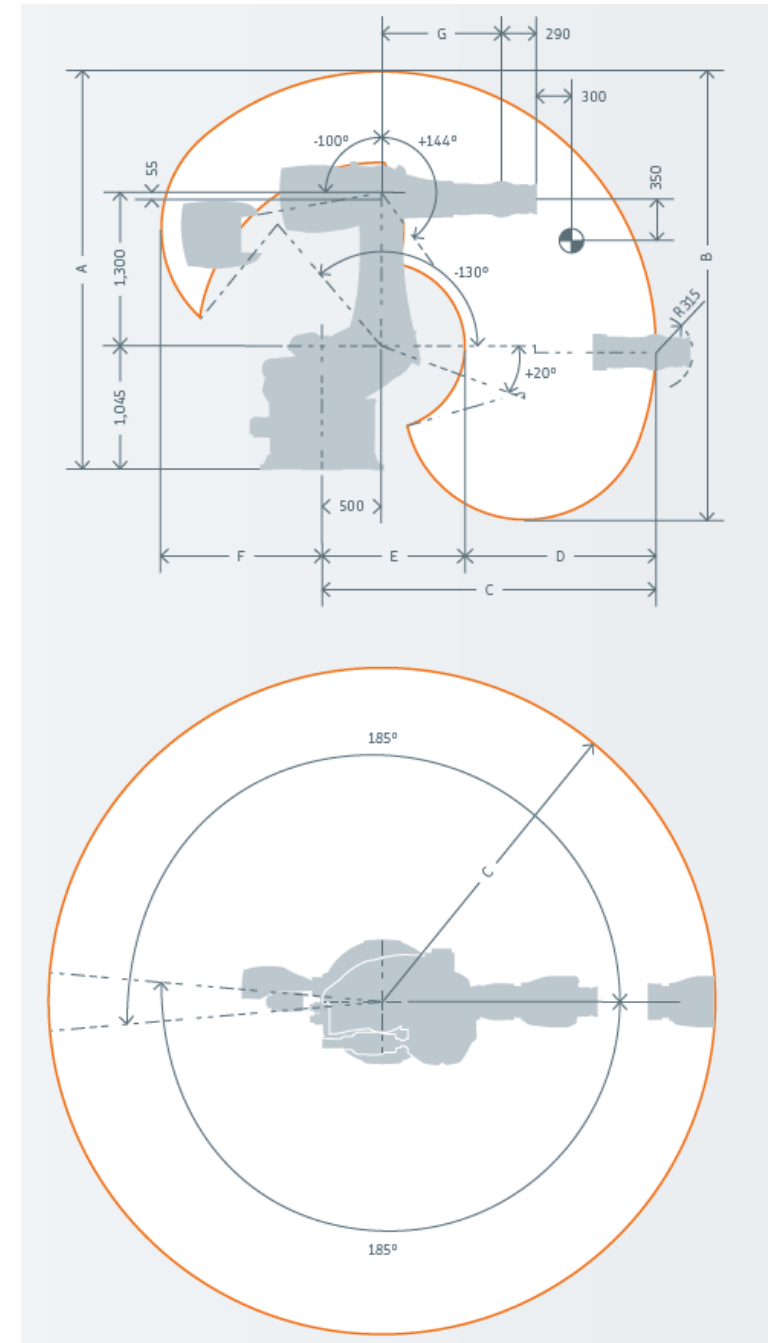
Workspace

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Workspace – Definition

Workspace –

The volume of space that the end-effector of the manipulator can reach



Workspace – Definition

Reachable Workspace –

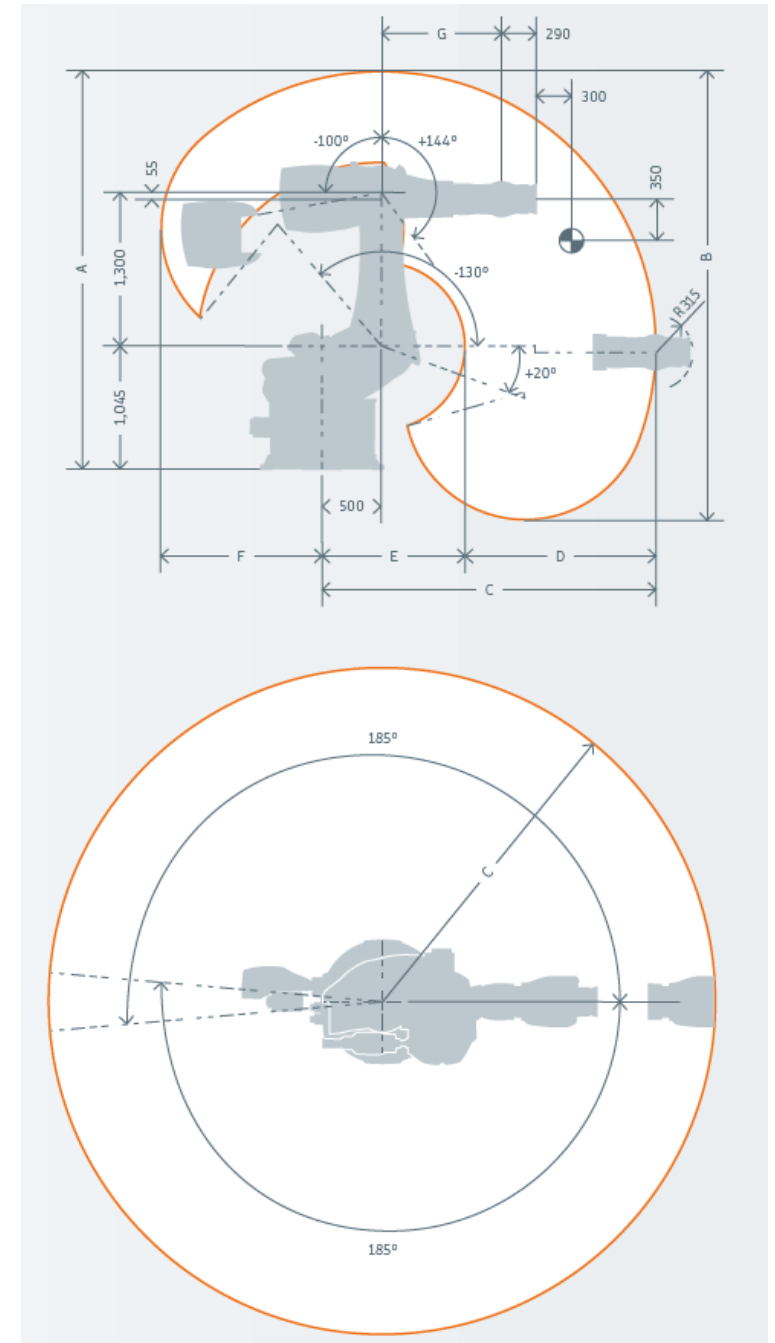
The volume of space that the end-effector of the manipulator can reach in **at least one** orientation

$$DW \subseteq RW$$

Dexterous Workspace –



The volume of space that the end-effector of the manipulator can reach in **all possible** orientations



Example – Planar RR Manipulator

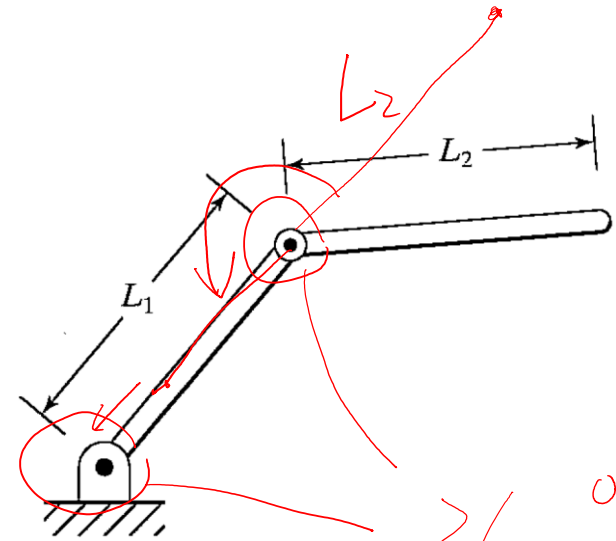
① $\underline{V_{max}} = L_1 + L_2$

② $\underline{V_{min}} = \begin{cases} L_1 - L_2 \\ 0 \\ L_2 - L_1 \end{cases}$

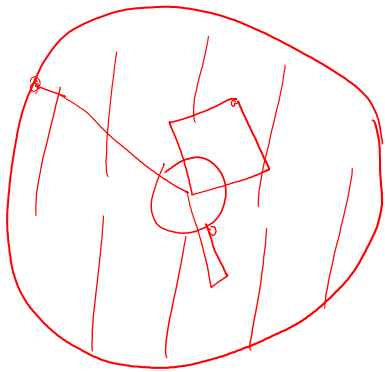
$L_1 > L_2$

$L_1 = L_2$

$L_1 < L_2$

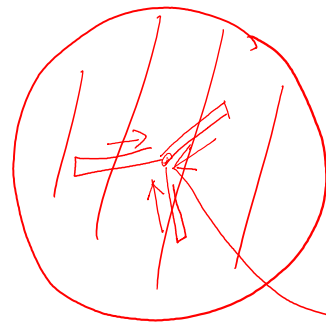


$\Rightarrow$



$L_1 \neq L_2$

No DW



$L_1 = L_2$

DW joint limit

Workspace Sketching

n D.o.F

Step 1 – Free the last joint n

Fix the other joints at boundaries

Draw the subworkspace^① resulting from the last joint

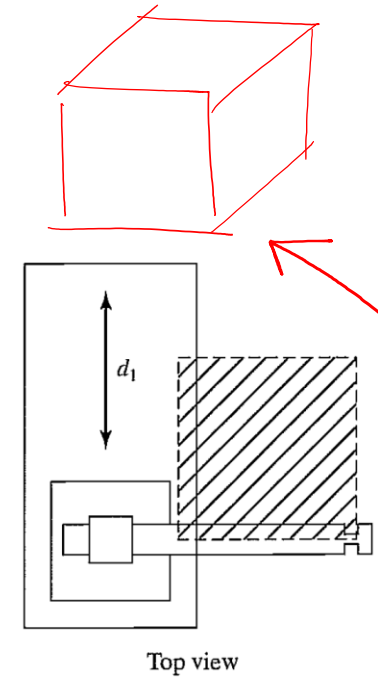
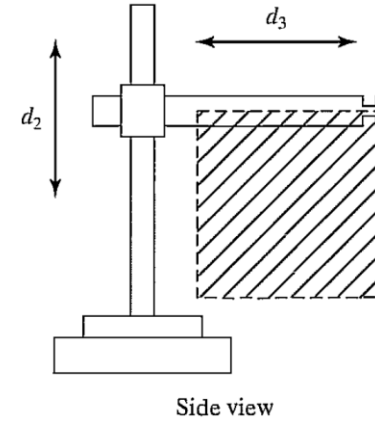
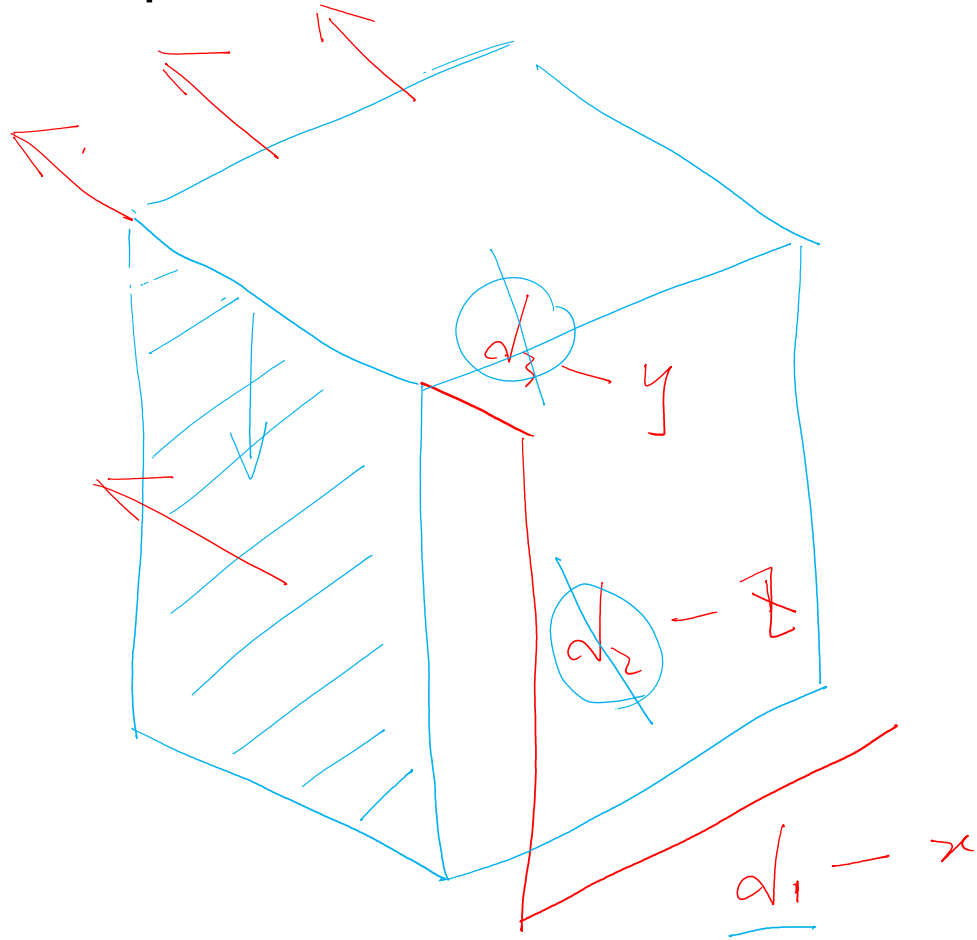
Step 2 – Ignore the last joint n ↘

Free the second to the last joint $n - 1$

Draw the subworkspace^② resulting from this joint based on the previous subworkspace

Step 3 – Iterate until the first joint 1

Example – Cartesian Manipulator



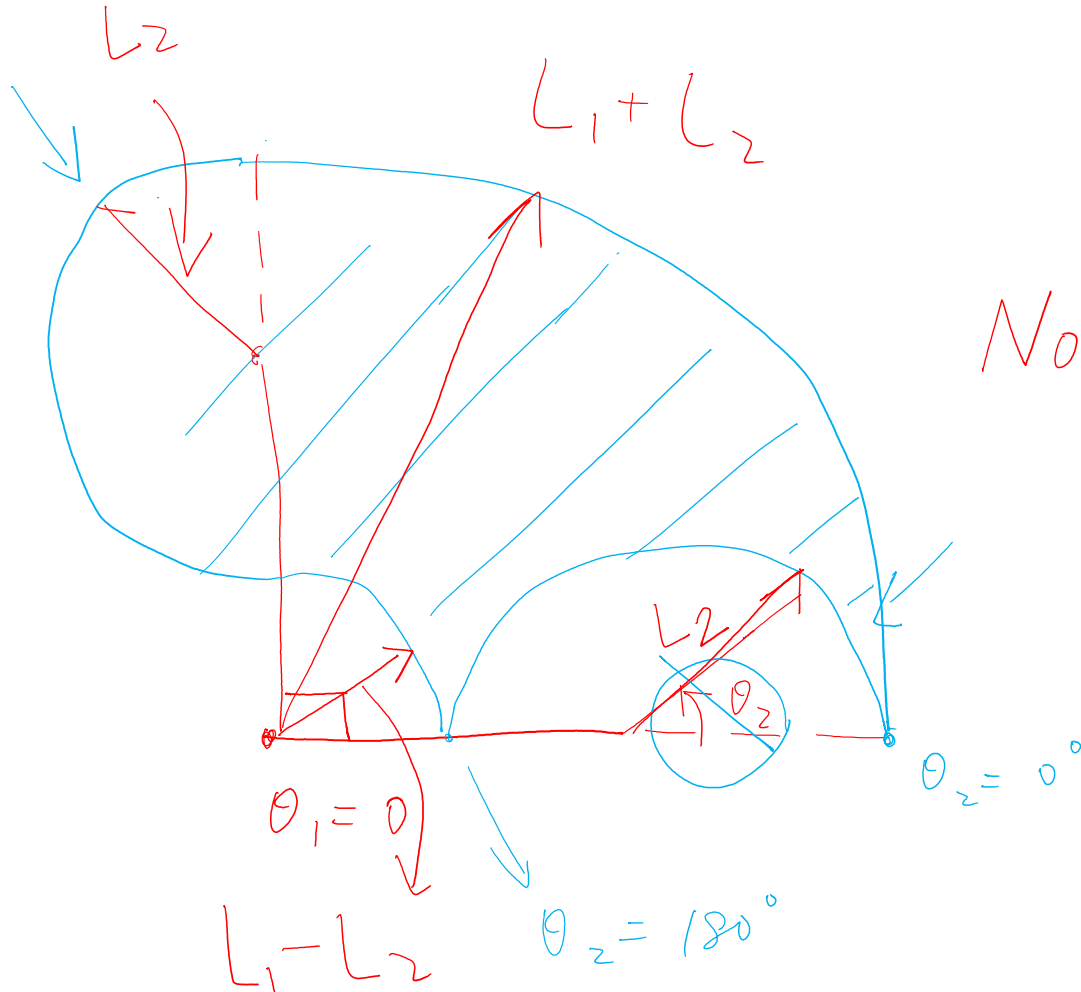
$$\underline{d_i} \leq d_i \leq \overline{d_i}$$

No DW

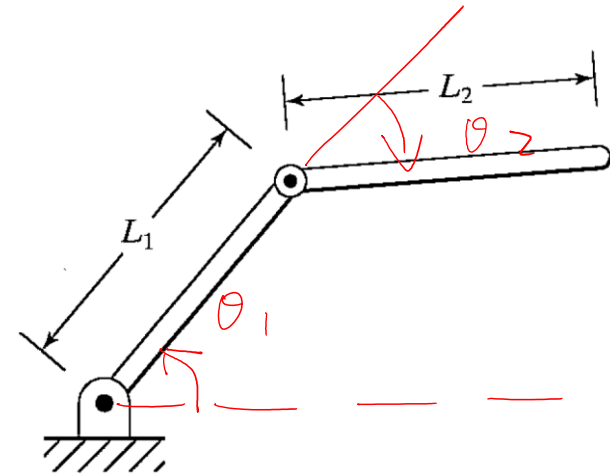
Example – Planar RR Manipulator

$$0 \leq \theta_1 \leq 90^\circ$$

$$0 \leq \theta_2 \leq 180^\circ$$



No DW

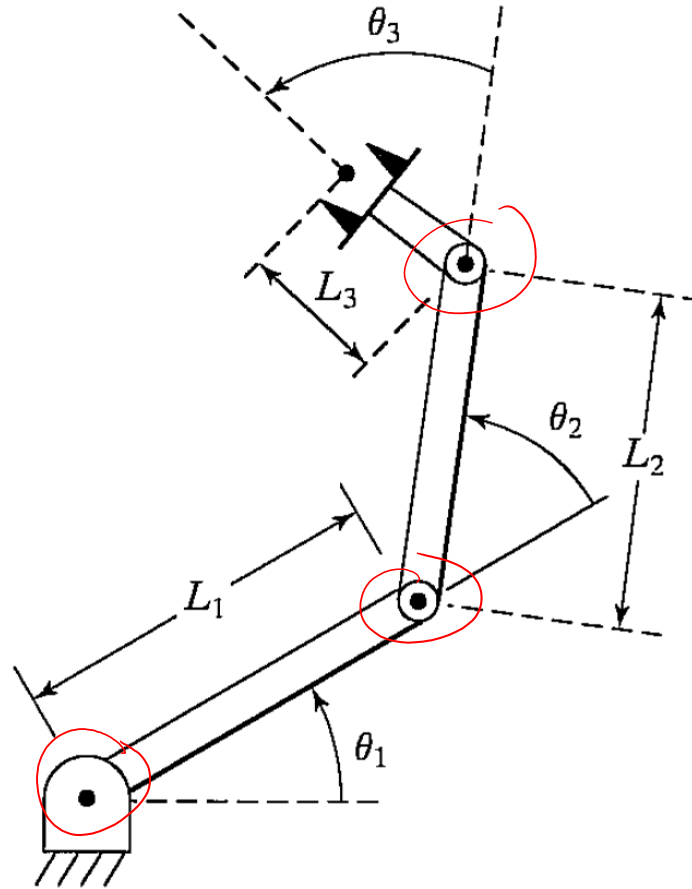


Exercise – Planar RRR Manipulator

$$L_1 = L_2 = L_3 ?$$

$$L_1 = L_2 + L_3 ?$$

...



Workspace Drawing

Forward Kinematics

Matlab

Inverse Kinematics
